

Measuring Diversity and Segregation with Convex Functions

Philip S. Chodrow *

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The study of quantitative measures for residential segregation extends far enough back that a review article in 1955 could express frustration and disappointment with the chaotic state of the methodology. Commenting then on the measures extant at the time, Duncan and Duncan [DD55] wrote:

1. *The proposed indexes of segregation have a number of hitherto unnoticed interrelationships which can be mathematically demonstrated.*
2. *Some of them have mathematical properties of which their proponents were unaware, and which lead to difficulties of interpretation.*
3. *As a consequence, the status of the empirical work already done with segregation indices is questionable, and their validity for further research is undermined.*

Over 70 years later, hints of this chaos remain. This author is aware of two instances in the last ten years of a segregation measure being published as novel despite already existing in the literature under a different name.

This article is not a systematic review of the literature, but rather a guided tour through one particular perspective on the problem of measuring segregation. This perspective begins with a general class of measures of diversity and segregation which

0.1

Broad motivation for the article

- Folks in the math community are turning to quantitative measures of segregation [DMW23, TBK⁺25, HSS25].

*Phil is an assistant professor of computer science at Middlebury College. His email address is pchodrow@middlebury.edu.

- There’s actually a pretty interesting and rich literature in quantitative sociology on this set of topics [Rea, RF02], going back all the way at least to [Sim49].
- MAUP is a thing [Ope84], and we’d like measures that behave in predictable ways as we change analytical scale.
- If you start engaging with these ideas systematically and mathematically, then you come naturally to convexity and Bregman divergences, which are what I am here to sell you today.

Figure 1 is a simple checkerboard pattern. What if we actually oriented the entire piece around the checkerboard¹:

Preliminaries

To make these ideas concrete, we formalize the idea of a *population* and measurements which reflect the diversity or segregation of that population. Our framework for defining populations is intentionally discrete, in recognition of the fact that all Census-type data sets provide demographic information in discrete areal units such as counties, tracts, or blockgroups.

Definition 1. For $\ell \in \mathbb{N}$, the *probability simplex* Δ_ℓ is the set

$$\Delta_\ell = \left\{ \mathbf{y} \in \mathbb{R}^\ell : \sum_{j=1}^{\ell} y_j = 1, y_j \geq 0 \text{ for all } j \in [\ell] \right\}. \quad (1)$$

¹The problem of constructing segregation measures that distinguish between panels (c) and (d) in Figure 1 is often called the “checkerboard problem.”

of probability vectors on ℓ elements.²

Definition 2. A population $\mathcal{P}$ with n locations in d spatial dimensions and ℓ demographic categories is a triple $\mathcal{P} = (\mathcal{X}, \mathbf{Y}, \mu)$, where:

- $\mathcal{X}$ is a finite metric space with n elements labeled x_i , $i = 1, \dots, n$.
- $\mathbf{Y} \in \mathbb{R}^{n \times \ell}$ is a row-stochastic matrix: for each row i , $\mathbf{y}_i \in \Delta_\ell$.
- $\mu \in \Delta_n$.

We denote the set of all populations with n locations, d dimensions, and ℓ demographic categories as $\mathcal{P}_{nd\ell}$.

In this definition, the matrix $\mathcal{X}$ represents space: element x_i of $\mathcal{X}$ can be viewed as giving the location of a population unit i . The matrix $\mathbf{Y}$ represents the demographic distribution of the population according to some attribute such as gender, race, or education level: the i th row $\mathbf{y}_i$ of $\mathbf{Y}$ gives the proportion of individuals in each demographic category at location i . Finally, the probability vector μ represents the population density across locations; the proportion of individuals living in location i is μ_i . The probability that a single individual, chosen uniformly at random from the population, lives in location i and is in demographic category j is $\mu_i y_{ij}$. The mean population vector $\bar{\mathbf{y}} = \sum_{i=1}^n \mu_i \mathbf{y}_i$ has entries giving the marginal probabilities that an individual chosen uniformly at random belongs to each demographic category.

Definition 3. A population measurement is a function $f : \mathcal{P}_{nd\ell} \rightarrow \mathbb{R}$.³

An example of a population measurement is $f(\mathcal{P}) = \sum_{i=1}^n \mu_i^2$, which measures the degree to which the population is concentrated in space. For fixed n , the measure f is maximized (with value $f(\mathcal{P}) = 1$) when the entire population resides in a single location x_i and minimized (with value $f(\mathcal{P}) = \frac{1}{n}$) when the population is uniformly distributed across all locations.

²Although it is sometimes common to refer to Δ_ℓ as an $(\ell - 1)$ -simplex (this being its manifold dimension), we write Δ_ℓ to emphasize its role as the set of probability distributions over ℓ categories.

³The function η could reasonably be described as a *population measure*, which is an entirely disjoint concept from a population measurement. We will avoid the phrase “population measure” and emphasize “measurement” whenever possible.

Diversity and Convexity

Heuristically, we often say that a population is *diverse* with respect to an attribute when it contains many categories of that attribute in roughly equal proportions. Diversity is usually considered to be a property of a population as a whole, without regard for spatial or demographic subdivisions. In our formal structure from Definitions 2 and 3, a diversity measure $f : \mathcal{P}_{nd\ell} \rightarrow \mathbb{R}$ is a function of the mean population demographic vector: $f(\mathcal{P}) = \psi(\bar{\mathbf{y}})$ for some function $\psi : \Delta_\ell \rightarrow \mathbb{R}$. It remains to specify the properties that ψ should have to capture an intuitive notion of diversity.

Working in mathematical ecology, [Pie77] gives a proposed set of criteria. For $\ell \in \mathbb{N}$ let $[\ell] = \{1, 2, \dots, \ell\}$. Let $\mathbf{e}_I \in \Delta_\ell$ be the vector with 1 in the indices specified by the multiindex $I \subseteq [\ell]$ and 0 elsewhere.

1. **Maximized by uniform.** “Diversity should be maximized when all groups are present in equal proportions.” In our setup, this means that $\frac{1}{\ell} \mathbf{e}_{[\ell]} = \operatorname{argmax}_{\mathbf{y} \in \Delta_\ell} \psi(\mathbf{y})$.
2. **Monotone under group addition.** “In two populations in which all groups are equally represented, the population with the greatest number of groups will be more diverse.” Mathematically, if I is a strict subset of $[\ell]$, then $\psi(\mathbf{e}_I / |I|) < \psi(\mathbf{e}_{[\ell]} / \ell)$.
3. **Additive under independent attributes.** “In a multiple classification (two or more qualitative variates) where the classifications are independent, the total diversity should equal the sum of the respective diversities.” This statement can be formalized as follows, interpreting “independent” in the sense of probability distributions: suppose that, for a given population $\mathcal{P} = (\mathcal{X}, \mathbf{Y}, \mu) \in \mathcal{P}_{nd\ell}$, there exist two other populations $\mathcal{P}_1 = (\mathcal{X}, \mathbf{U} \in \mathbb{R}^{n \times \ell_1}, \mu)$ and $\mathcal{P}_2 = (\mathcal{X}, \mathbf{V} \in \mathbb{R}^{n \times \ell_2}, \mu)$ such that $\ell = \ell_1 \ell_2$ and a bijection $\alpha : [\ell_1] \times [\ell_2] \rightarrow [\ell]$ such that $y_{ij} = u_{i\alpha_1(j)} v_{i\alpha_2(j)}$ for all $i \in [n]$ and $j \in [\ell]$. Then $f(\mathcal{P}) = f(\mathcal{P}_1) + f(\mathcal{P}_2)$.

We will soon find that properties (1) and (2) are satisfied by a broad range of functions ψ , while property (3) is more restrictive and essentially characterizes a single diversity measure.

Segregation

Spatial Scale

Extensions

Figures we still want

- Side-by-side maps showing local information and maybe a regionalization.
- Possibly: scatterplot comparing local information and mutual information.

[Cho25] [Cho17]

[BMDG05]

References

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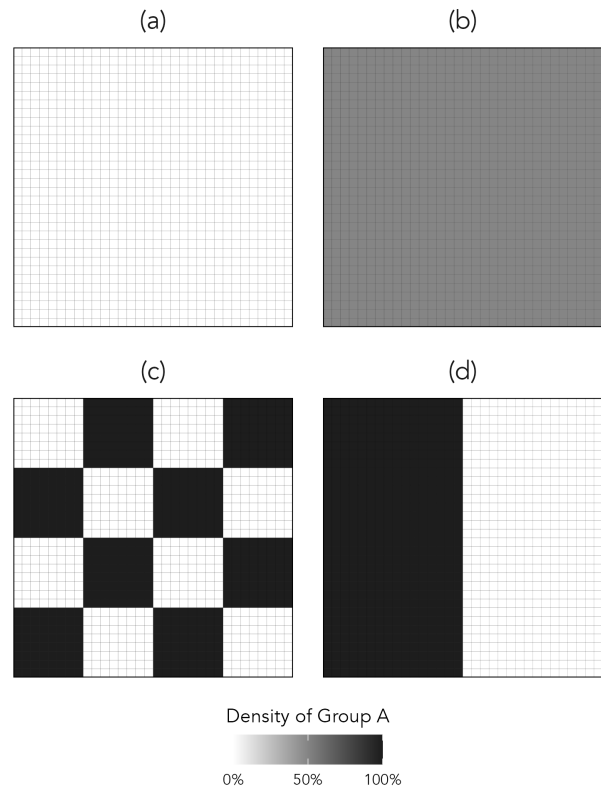


Figure 1: A simple checkerboard pattern. Each of the four panels is composed of 32 cells. Panel (c) is further arranged into 4 blocks of 8 cells each.

[TBK⁺25] Chad M Topaz, Heather Z Brooks, Unchitta Kan, Björn Sandstede, and Christian Michael Smith, *Diversity, identity, and data*, The American Mathematical Monthly **132** (2025), no. 1, 36–47.

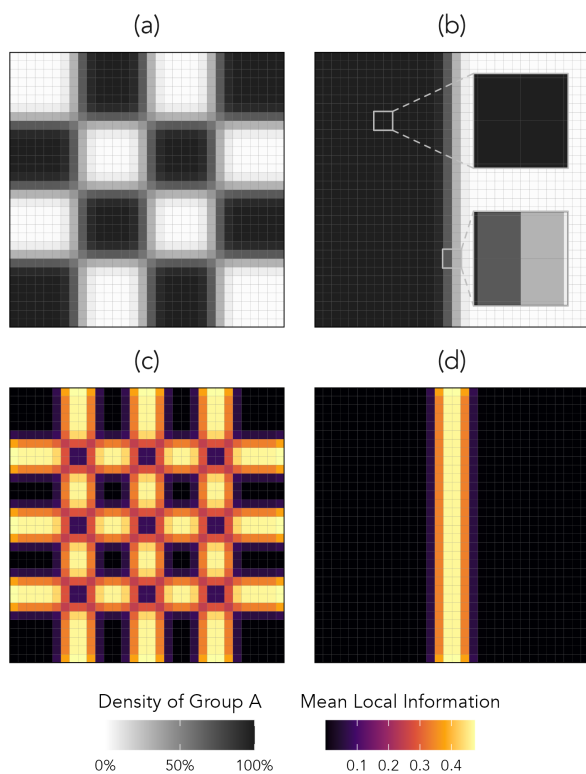


Figure 2: The smoothing bandwidth is 1 pixels. The mean local information of the checkerboard pattern is 0.228, while the mean local information of the segregated pattern is 0.054.

Figure 3: Don't forget that we need to do some empirical stuff.